PRELIMINARY EXPERT RESEARCH MEMORANDUM / REVIEW COPY

Cogent Insights Licensing Inc. v. HMS Industrial Networks AB

U.S. Patent No. 9,794,797 game-theoretic and synthetic-economy management of ad hoc wireless-network resources, alleged against HMS Industrial Networks' unspecified networking products; the Exhibit 2 mapping was unavailable.

Independent preliminary research based on public information.

FILED	FORUM	DOCKET	MATTER
2026-08-20	E.D. Tex.	2:26-cv-00720	Patent Infringement

Prepared 2026-08-25. No attorney or expert contacted. Availability, interest, compensation, and conflicts not checked.

Public filing, likely recipient, and technical frame

CAPTION	Cogent Insights Licensing Inc. v. HMS Industrial Networks AB
FORUM / DOCKET	E.D. Tex. / 2:26-cv-00720
FILED / TYPE	2026-08-20 / patent infringement
PUBLIC DOCKET	Open docket

VERIFIED PUBLIC RECORD

The public docket verifies that Cogent Insights Licensing Inc. v. HMS Industrial Networks AB was filed on 2026-08-20 in E.D. Tex. under docket 2:26-cv-00720.

LIKELY RESEARCH PURCHASER

Isaac Rabicoff - Outside counsel, Rabicoff Law LLC. For each qualified matter, the individual public docket identifies Isaac Rabicoff as counsel appearing for Cogent Insights Licensing Inc. and as filer of the complaint or opening papers. Treating him as the likely purchaser is an analytical inference from that role, not a claim about internal responsibility.

isaac@rabilaw.com - publicly listed on the official firm biography; not inferred.

TECHNICAL FRAME

U.S. Patent No. 9,794,797 game-theoretic and synthetic-economy management of ad hoc wireless-network resources, alleged against HMS Industrial Networks' unspecified networking products; the Exhibit 2 mapping was unavailable.

Secondary-source limitation: The reusable bench applies only to the E.D. Texas matters asserting U.S. Patent No. 9,794,797. It must not be reused for Cogent Insights Licensing Inc. v. Caterpillar Inc., which concerns a different induction-generator patent. The public complaint analyses report that referenced Exhibit 2 claim charts were unavailable. The public record reviewed does not identify specific accused models, asserted claims by defendant, product configurations, or an element-by-element infringement theory. Docket 2:26-cv-00718 remains outside the qualified set because the campaign's separate qualification shard did not locate an individual public docket-detail page verifying counsel's role. Open public synopsis

REPORTED ASSERTED PATENTS

U.S. Patent No. 9,794,797

Questions likely to require expert input

This issue map is a research plan, not a legal conclusion. The filed complaint, patent exhibits, intrinsic record, and discovery must control.

01 What would a person of ordinary skill have understood by 'automated negotiation,' 'game theoretic decision-making,' 'self-organize,' 'cooperative agent,' and 'strategic value' in the 2005 priority-period context?

A communications-networking expert paired with a game-theory or mechanism-design expert can identify the technical meaning, conventional usage, and relationships among these terms without deciding the legal construction.

Potential evidence: Patent specification, prosecution history, and asserted claims; Pre-October 2005 literature on ad hoc networks, congestion pricing, VCG and Kelly mechanisms, auctions, and distributed resource allocation; Contemporaneous standards and protocol descriptions

02 Does an accused product actually conduct a negotiation with a remote wireless device, and is the operative decision process game-theoretic rather than ordinary rule-based control, optimization, scheduling, or protocol exchange?

The expert will likely need to distinguish a modeled strategic interaction among agents from ordinary deterministic configuration, feedback control, channel selection, beam management, or load balancing.

Potential evidence: Firmware and source code; Protocol traces, message formats, configuration files, and state machines; Architecture documents and developer testimony; Reproducible product testing

03 Does the accused antenna system have an alterable directional vector with at least two states having different spatial characteristics and different potential interference, and is the selected state dependent on the claimed negotiation?

This requires RF and antenna-system analysis of beamforming or directional states, interference estimation, control causality, and the relationship between software decisions and the radiated pattern.

Potential evidence: Antenna schematics, array geometry, beam tables, calibration data, and RF specifications; Beam-management or steering code and logs; Anechoic-chamber, over-the-air, or controlled interference testing; Product manuals and regulatory filings

04 Which accused product configurations, optional features, network roles, and deployment conditions could satisfy every asserted technical limitation?

Because the current public record does not identify models or provide the referenced claim charts, a product-by-product and configuration-specific analysis is essential; a feature being theoretically supported does not establish that it is enabled, used, or causally connected in the claimed combination.

Potential evidence: Accused-product identification and sales configurations; Feature licenses, defaults, release notes, and customer documentation; Source-code branches and version histories; Network topology and operational telemetry

Questions likely to require expert input

05 What pre-priority systems and publications disclosed the claimed combination or its components, including game-theoretic resource allocation, VCG or auction mechanisms, multihop ad hoc networking, beam steering, interference-aware control, and distributed negotiation?

A complementary game-theory/network-optimization and wireless-systems team can build a dated technical baseline, assess combinations, and identify whether the patent applies known economic mechanisms to known wireless control elements.

Potential evidence: IEEE, ACM, INFORMS, and operations-research publications before October 4, 2005; Wireless and ad hoc networking standards, textbooks, theses, and technical reports; Prior patents and working systems; Citation and authorship records capable of supporting authentication and technical explanation

06 Does the specification teach a skilled person how to implement the full claimed breadth without undue experimentation, particularly the negotiation, game-theoretic objective, interference comparison, and directional-antenna control loop?

The expert can assess the disclosure's algorithms, inputs, objective functions, convergence assumptions, signaling requirements, antenna implementation detail, and reproducibility across the claimed scope, while leaving the ultimate legal conclusions to counsel and the court.

Potential evidence: Patent examples, figures, algorithms, and incorporated references; Contemporaneous implementation complexity and computing constraints; Simulation or prototype requirements; Ordinary skill and common general knowledge at the priority date

07 What technical value, if any, is specifically attributable to the claimed game-theoretic negotiation plus directional-antenna linkage rather than to conventional radios, processors, network protocols, or unclaimed product features?

A technical expert can identify the smallest relevant technical unit, available noninfringing alternatives, feature usage, and performance effects to support counsel's damages and apportionment work without offering an economic damages opinion unless separately qualified.

Potential evidence: Feature-level performance testing and telemetry; Design alternatives and implementation cost records; Marketing and customer-use evidence; Technical contribution of asserted limitations relative to the full product

Candidates 1-2 for further diligence

Ranked for preliminary research priority based on subject-matter fit, relevant public work, testimony, and visible diligence burden. This is not a retention recommendation.

01 Ramesh Johari

Professor of Management Science and Engineering and, by courtesy, of Electrical Engineering and Computer Science, Stanford University

TECHNICAL FIT

Exceptional fit for the patent's core game-theoretic network-resource question. His work directly addresses efficiency loss in network resource-allocation games, congestion pricing, network formation, traffic engineering, and mechanism design.

RELEVANT WORK

Coauthor of 'Efficiency loss in a network resource allocation game,' Mathematics of Operations Research 29(3), 2004—squarely within the patent's priority-period technical vocabulary. Published on communication requirements of VCG-like mechanisms, routing and peering, congestion games, and game-theoretic resource allocation. Stanford identifies him as a professor across management science, electrical engineering, and computer science.

PUBLIC LITIGATION EXPERIENCE

No candidate-specific public expert report, declaration, deposition, or trial-testimony record was located in the bounded search. This is not evidence of no litigation experience.

POINTS FOR FURTHER DILIGENCE

His most direct fit is the economic and algorithmic meaning of network games, not RF antenna-array design, beam-pattern measurement, or product teardown. His current research also extends beyond communications networks into online platforms.

Source 1 | Source 2 | Source 3

Draft email | rjohari@stanford.edu | public email source

02 Bruce Hajek

Hoeft Chair and Professor of Electrical and Computer Engineering; Research Professor, Coordinated Science Laboratory, University of Illinois Urbana-Champaign

TECHNICAL FIT

Unusually close combination of communication networks, auction theory, game theory and mechanism design, stochastic analysis, combinatorial optimization, routing, congestion pricing, and fair resource allocation.

RELEVANT WORK

Published on VCG-Kelly mechanisms for allocation of divisible goods, network pricing, combinatorial auctions for wireless spectrum, routing, load balancing, and random multiple access. Illinois describes his research interests as game theory and mechanism design, optimization, wireless communication, and stochastic analysis. His communication-network work includes fair distribution of network resources among competing information flows.

PUBLIC LITIGATION EXPERIENCE

No candidate-specific public expert report, declaration, deposition, or trial-testimony record was located in the bounded search. This is not evidence of no litigation experience.

POINTS FOR FURTHER DILIGENCE

His work is principally theoretical and analytical; a separate RF implementation expert may be needed for accused beamforming hardware, firmware, and over-the-air testing. Institutional and research relationships require ordinary conflicts diligence.

Source 1 | Source 2 | Source 3

Draft email | b-hajek@illinois.edu | public email source

Candidates 3-4 for further diligence

03 Asu Ozdaglar

MathWorks Professor of Electrical Engineering and Computer Science; Head of MIT EECS; Deputy Dean of Academics, MIT Schwarzman College of Computing

TECHNICAL FIT

Direct fit for the intersection the patent invokes: game theory, pricing and resource-allocation games, network economics, distributed optimization, and wireless and wireline network control.

RELEVANT WORK

Coauthor of 'Network Games: Theory, Models, and Dynamics' and a chapter on incentives and pricing in communication networks. Published on distributed cross-layer algorithms for optimal control of multihop wireless networks and iterative auction design. MIT identifies her research expertise in optimization, game theory, networks, and distributed algorithms for network resource allocation.

PUBLIC LITIGATION EXPERIENCE

No candidate-specific public expert report, declaration, deposition, or trial-testimony record was located in the bounded search. This is not evidence of no litigation experience.

POINTS FOR FURTHER DILIGENCE

Her recent public research emphasis includes AI and social/economic networks, and her substantial administrative duties may affect practical scheduling. She is not primarily an antenna-measurement or product-reverse-engineering specialist.

Source 1 | Source 2 | Source 3

Draft email | asuman@mit.edu | public email source

04 Rayadurgam Srikant

Fredric G. and Elizabeth H. Nearing Endowed Professor of Electrical and Computer Engineering, University of Illinois Urbana-Champaign; Co-Director, C3.ai Digital Transformation Institute

TECHNICAL FIT

Strong bridge between mathematical network optimization and actual communication-network control, including congestion, stochastic control, pricing, and distributed resource allocation.

RELEVANT WORK

Coauthor of 'Network Optimization and Control' and author of 'The Mathematics of Internet Congestion Control.' Coauthored the communication-networks chapter on incentives and pricing with Asu Ozdaglar. Illinois lists communication networks, stochastic control, and applied probability among his research interests.

PUBLIC LITIGATION EXPERIENCE

No candidate-specific public expert report, declaration, deposition, or trial-testimony record was located in the bounded search. Searches were disambiguated from other professionals named Srikant. This is not evidence of no litigation experience.

POINTS FOR FURTHER DILIGENCE

The public record emphasizes theory, stochastic control, and network algorithms rather than directional-antenna hardware or RF chamber testing. Current work also includes machine learning beyond the asserted patent's 2005 context.

Source 1 | Source 2 | Source 3

Draft email | rsrikant@illinois.edu | public email source

Candidates 5-6 for further diligence

05 Steven H. Low

Frank J. Gilloon Professor of Computing and Mathematical Sciences and Electrical Engineering, California Institute of Technology; Director, Schmidt Academy for Software Engineering

TECHNICAL FIT

Seminal fit for network utility maximization, distributed congestion control, network architecture, and the relationship between local algorithms, equilibrium, constraints, and system-wide resource allocation.

RELEVANT WORK

Caltech recognizes his theoretical foundations and real-world deployment of Internet congestion control and optimization. His network-control text develops network utility maximization through convex optimization, primal-dual algorithms, routing matrices, link capacity, and equilibrium. He teaches Control and Optimization of Networks and works on network architecture and energy-efficient networking.

PUBLIC LITIGATION EXPERIENCE

No candidate-specific public expert report, declaration, deposition, or trial-testimony record was located in the bounded search. Searches were disambiguated from other people named Steven Low. This is not evidence of no litigation experience.

POINTS FOR FURTHER DILIGENCE

His closest work concerns utility maximization and congestion-control architecture rather than VCG auctions or antenna-direction selection. His recent work includes power systems, so a wireless RF specialist would still be necessary.

Source 1 | Source 2 | Source 3

Draft email | slow@caltech.edu | public email source

06 Ness B. Shroff

Ohio Eminent Scholar in Networking and Communications and Professor of Electrical and Computer Engineering and Computer Science and Engineering, The Ohio State University

TECHNICAL FIT

Broad, current fit across network optimization, stochastic control, algorithm design, pricing, security, wireless systems, and communication, computing, storage, and cloud networks.

RELEVANT WORK

Ohio State describes his work as contributing to the design, control, performance, pricing, and security of communication and computing networks. His public profile specifically identifies network optimization, stochastic control, and algorithmic design as core research areas. Former Editor-in-Chief of IEEE/ACM Transactions on Networking and a senior leader in the networking research community.

PUBLIC LITIGATION EXPERIENCE

No candidate-specific public expert report, declaration, deposition, or trial-testimony record was located in the bounded search. This is not evidence of no litigation experience.

POINTS FOR FURTHER DILIGENCE

His public profile is broad and does not itself establish hands-on experience with the particular accused devices, antenna arrays, or firmware. The closest claim issues may require a narrower mechanism-design or RF specialist alongside him.

Source 1

Draft email | shroff.11@osu.edu | public email source

Candidates 7-8 for further diligence

07 Eytan H. Modiano

Richard Cockburn Maclaurin Professor in Aeronautics and Astronautics, Massachusetts Institute of Technology; Director, Communications and Networking Research Group

TECHNICAL FIT

Direct communications-network expertise in network optimization, routing, scheduling, queueing, graph algorithms, cross-layer control, wireless networks, and robust architectures.

RELEVANT WORK

Published on network utility maximization, optimal control for generalized network-flow problems, robust network design, spectrum sharing, and multihop wireless routing and scheduling. MIT describes his research as spanning network optimization, queueing, graph theory, protocols and algorithms, and physical-layer communications. His work includes cost-effective, scalable, and robust architectures across wireless, satellite, and optical networks.

PUBLIC LITIGATION EXPERIENCE

No candidate-specific public expert report, declaration, deposition, or trial-testimony record was located in the bounded search. This is not evidence of no litigation experience.

POINTS FOR FURTHER DILIGENCE

His strongest public work concerns optimization, routing, scheduling, and robustness, not auction incentives or synthetic-currency mechanisms. Product-level directional-antenna testing may require a separate RF engineer.

Source 1 | Source 2 | Source 3 | Source 4

Draft email | modiano@mit.edu | public email source

08 Jean Walrand

Professor Emeritus of Electrical Engineering and Computer Sciences, University of California, Berkeley

TECHNICAL FIT

Deep fit for communication-network architecture, queueing, scheduling, congestion control, resource sharing, stochastic systems, and the interface between networking theory and implementable protocol behavior.

RELEVANT WORK

Coauthor of books on High-Performance Communication Networks, Scheduling and Congestion Control, and Sharing Network Resources. Named inventor on patents involving network power management, switch fabrics, and provisional IP-aware virtual paths. His communication-network textbook has appeared as technical prior-art literature in a public PTAB petition, although that does not mean he served as the retained expert.

PUBLIC LITIGATION EXPERIENCE

A public PTAB petition cites Walrand and Varaiya's textbook as an exhibit, but the petition identifies a different declarant. No candidate-specific expert role was located in the bounded search.

POINTS FOR FURTHER DILIGENCE

He is emeritus, and public evidence does not establish current hands-on work with the accused products. His breadth across network theory may need supplementation by a modern cellular, Wi-Fi, industrial-wireless, or antenna-array specialist depending on the accused instrumentality.

Source 1 | Source 2 | Source 3

Draft email | wlr@eecs.berkeley.edu | public email source

Candidates 9-10 for further diligence

09 John N. Tsitsiklis

Clarence J. LeBel Professor of Electrical Engineering and Computer Science, Massachusetts Institute of Technology

TECHNICAL FIT

Strong foundational fit for optimization, network resource allocation, network flows, decentralized computation, decision-making in networks, and the efficiency of strategic resource allocation.

RELEVANT WORK

Coauthored the priority-period paper 'Efficiency loss in a network resource allocation game' with Ramesh Johari. MIT lists network resource allocation and decision-making in networks among his expertise. Coauthor of a linear-optimization text covering network-flow and discrete-optimization problems and of foundational work on parallel and distributed computation.

PUBLIC LITIGATION EXPERIENCE

No candidate-specific public expert report, declaration, deposition, or trial-testimony record was located in the bounded search. This is not evidence of no litigation experience.

POINTS FOR FURTHER DILIGENCE

His public work is highly mathematical and foundational. He is less directly identified with antenna systems, RF interference measurement, or product firmware, and may be best suited to validity, ordinary-skill, and algorithmic-meaning questions.

Source 1 | Source 2

Draft email | jnt@mit.edu | public email source

10 David N. C. Tse

Thomas Kailath and Guanghan Xu Professor of Engineering and Professor of Electrical Engineering, Stanford University

TECHNICAL FIT

The bench's strongest complementary physical-layer and multiuser-wireless candidate. His work on wireless communication, proportional-fair scheduling, information theory, path diversity, throughput, and multiuser systems can test whether claimed economic language maps to real radio behavior.

RELEVANT WORK

Coauthor of the widely used text 'Fundamentals of Wireless Communication.' Inventor of the proportional-fair scheduling algorithm used in cellular systems, according to Stanford's profile. Named inventor on patents involving throughput, multiple-mobile-station transmission, path diversity, and bit-rate services.

PUBLIC LITIGATION EXPERIENCE

No candidate-specific public expert report, declaration, deposition, or trial-testimony record was located in the bounded search. Searches were disambiguated from other professionals named David Tse. This is not evidence of no litigation experience.

POINTS FOR FURTHER DILIGENCE

His best fit is wireless-system operation, multiuser resource allocation, and technical apportionment—not auction theory or synthetic economic value. His current research also spans areas outside wireless networking, so the specific engagement scope would need careful definition.

Source 1 | Source 2

Draft email | dntse@stanford.edu | public email source

Preliminary candidates; no conflicts or availability determination

- Run party, affiliate, counsel, inventor, funder, and technology conflicts.

- Confirm testimony history, exclusions, compensation, publications, patents, and deposition performance.

- Investigate licensing, grants, consulting, commercial interests, and prior technical positions.

- Confirm role fit, availability, and interest only after counsel authorizes outreach.

ONGOING MATTER-SPECIFIC RESEARCH

Flint Witness Research works in the post-filing window, turning pleadings and public technical records into issue maps, researched candidate slates, and diligence starting points as the case develops.

Schedule a 15-minute conversation

CORE SOURCES

Public docket: https://dockets.justia.com/docket/texas/txedce/2%3A2026cv00720/248566

Official attorney biography and email: https://ptacts.uspto.gov/ptacts/public-informations/petitions/1558005/download-documents?artifactId=ENAiVyKmiqCh4dFsoask1gMdS4vzuY0DPM3HVkUQ5P2SnlztfEvA21E

Public technical context source: https://ai-lab.exparte.com/case/dct/txed/2%3A26-cv-00720/doc/analysis/1

LIMITATIONS

The reusable bench applies only to the E.D. Texas matters asserting U.S. Patent No. 9,794,797. It must not be reused for Cogent Insights Licensing Inc. v. Caterpillar Inc., which concerns a different induction-generator patent. The public complaint analyses report that referenced Exhibit 2 claim charts were unavailable. The public record reviewed does not identify specific accused models, asserted claims by defendant, product configurations, or an element-by-element infringement theory. Docket 2:26-cv-00718 remains outside the qualified set because the campaign's separate qualification shard did not locate an individual public docket-detail page verifying counsel's role.

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